



Simulating the Development of Instinct via Synthetic Genomes
and Evolutionary Algorithms

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Abstract

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Abstract

Instinct represents a fundamental form of behavior expressed by living organisms without prior learning, transmitted genetically across generations to enhance survival in diverse environments. This study develops a computational model of “instinct for survival” by encoding genetic information and simulating evolutionary processes. Each virtual organism carries a DNA-like code that decodes into a gene regulatory network, functioning as a brain to control traits and behaviors. The model emphasizes zero-shot and near-zero-shot conditions, enabling newly generated organisms to demonstrate instinctive behaviors immediately, even in novel environments. By applying principles of natural selection, the framework highlights the evolution and transmission of traits and behaviors across generations. Beyond biology, the research extends to robotics and artificial intelligence, where systems can be designed with innate behavioral rules to adapt and respond flexibly without extensive training data. This work contributes to computational evolution and artificial life, offering pathways for designing adaptive AI and robotic systems that more closely mirror the autonomy and resilience of living organisms.

Keywords: Artificial life, Evolution computational, Natural selection, Evolution algorithm, Gene Regulation Network, Baldwin effect, Plasticity

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Chapter 1 Introduction

1.1 Rational

Instinct is a fundamental behavior that living beings can express without learning. It is a mechanism transmitted genetically from one generation to the next. For example, newly hatched crabs can immediately walk toward the sea. Such behaviors reflect the role of genetics in the determination of survival strategies in different environments. However, contemporary perspectives in developmental ethology suggest that instinct is not merely a static, hardwired program, but a robust outcome of developmental processes that rely on the interplay between genetic inheritance and consistent environmental contexts. Biologists and psychologists are therefore interested in studying instinct, as a key foundation for understanding life and evolution.

The concept of instinct has also inspired applications in robotics and artificial intelligence. Systems can be designed with 'basic rules' or pre-programmed reflexes, such as linking sensors with actuators to create automatic responses, for example stopping when an obstacle is detected. This is an example of how a robot or mechanism can demonstrate behavior similar to instinct, they are often rigid, manually designed, and lack the adaptability to work with unforeseen environment.

This research focuses to develop the computational model of 'instinct for survival' through genetic encoding and evolutionary processes. It draws on the principle of natural selection, which explains that genetic traits suitable for survival have a greater chance of being passed on to future generations and the Baldwin Effect, exploring how lifetime plasticity can guide the evolutionary process. In the proposed model, each virtual organism possesses a genetic code analogous to DNA, which is translated into a gene regulatory network (GRN) and this research incorporates mechanisms of phenotypic plasticity and epigenetic inheritance within the GRN controller. The GRN functions as a developmental controller, how the agent perceives environmental stimuli and executes actions throughout its lifetime.

The significance of this research is to demonstrating that behaviors can emerge and transmitted across generation through evolutionary mechanism, in a way that can be tested under zero-shot or near-zero-shot conditions. This means that newly born organisms can display instinctive behaviors immediately, even in environments slightly different from those of their ancestors. This approach investigates the hypothesis of 'genetic assimilation,' where learned behaviors eventually become innate instincts, testing the generalizability and robustness of evolved behavioral strategies.

This research has implications for both theoretical understanding and practical applications. Theoretically, it provides a computational framework for studying the

evolution of innate behaviors and the role of developmental processes in behavioral expression. Practically, it offers a pathway toward designing autonomous systems with built-in adaptive behaviors that do not require extensive training data, potentially enabling robots and AI agents to respond effectively to novel situations from initialization. By bridging computational evolution, artificial life, and adaptive behavior research, this work contributes to the broader goal of creating artificial systems that exhibit increasingly lifelike adaptation and development.

1.2 Problem statement

This research addresses the question: Can instinctive survival behaviors emerge through evolutionary processes and be expressed immediately in descendant organisms without prior individual learning?

The proposed approach develops a computational framework model of instinct that enables artificial organisms to exhibit survival. Models the evolution of developmental rules through a synthetic genome and a Gene Regulatory Network (GRN). This model integrates phenotypic plasticity and epigenetic inheritance, allowing the evolutionary process to leverage lifetime learning as a catalyst for adaptation. By simulating the Baldwin Effect, the system demonstrates how flexible behavioral strategies can be genetically assimilated over generations, resulting in robust survival instincts that function immediately upon initialization

1.3 Research Objectives

This research aim to develop a computational framework utilizing Gene Regulatory Networks (GRNs) as behavioral controllers for artificial agents. To simulate the emergence of instinctive survival strategies, by integrating evolutionary algorithms with lifetime synaptic plasticity via Hebbian learning, the study seeks to investigate the Baldwin Effect, specifically analyzing how individual learning acts as a catalyst for genetic assimilation to transform costly learned behaviors into efficient, innate instincts that exhibit zero-shot adaptation in dynamic environments. Furthermore, the research want to investigate evolutionary trajectories, thereby providing a detailed analysis of how complex behavioral patterns are encoded, preserved, and refined within the genetic regulatory architecture across generations.

1.3.1 Contribution of the research

1. Develops a simulation that models artificial life forms with genetic codes. This system enables dynamic visualization and analysis of instinctive behavior and form evolution across multiple generations.
2. Designs a computational framework for the encoding and decoding of Artificial DNA into Gene Regulatory Networks (GRNs). This framework facilitates the precise regulation of genetic information, supporting the targeted development of specific traits, phenotypic plasticity, and behavioral responses.
3. The research provides a computational validation of the Baldwin Effect, illustrating how lifetime learning (plasticity) acts as an evolutionary scaffold that eventually facilitates the emergence of robust, innate instincts.

4. A framework for artificial environments is designed to emulate natural selection and evolutionary dynamics. This environment supports self-adaptive and emergent behavior and morphological structure of organism, enabling experimental studies on artificial evolution and the interaction between genetic variation and environment.
5. The research investigates the capacity of artificial organisms to exhibit instinctive or adaptive behaviors under zero-shot and near-zero-shot learning conditions. This contribution demonstrates the potential of artificial life systems to generalize responses beyond prior training, paralleling certain aspects of biological cognition and intuition.

1.4 Scope of Research

1. The simulation is designed as a 2D waterworld environment, serving as the digital habitat for studying artificial life and evolutionary processes.
2. The artificial life forms in this research are protozoa-like entities, representing simple organisms capable of demonstrating emergent and instinctive behaviors.
3. The study assumes that behavioral strategies are indirectly encoded within a synthetic genome. This genetic information, which governs the Gene Regulatory Network (GRN), is transmitted and mutated across generations through simulated asexual reproduction.
4. In this research, 'instinct' is defined not merely as a repetition of behavior, but as a robust, innate capability to execute effective survival strategies immediately upon birth (Zero-shot), resulting from the evolutionary tuning of developmental parameters.
5. The evolutionary outcomes are validated under zero-shot and near-zero-shot conditions. This scope focuses on observing whether agents can generalize inherited strategies to environments that are identical or slightly varied from their ancestors, thereby confirming the genetic assimilation of learned behaviors.

1.5 Expected Benefit

1. A computational framework for encoding, encrypting, and decrypting artificial genetic codes and gene regulatory networks that can support the development of traits and behavior of living organism.
2. A Simulation program for visualizing instinctive traits and behaviors across generations.
3. A proof of concept that demonstrates how instinct and inheritance can be modeled within the fields of artificial life, evolutionary computation, and bio-inspired algorithms.

Chapter 2 Related Work

2.1 Theoretical Foundations of Instinct and Innate Behavior

The question of how complex innate behaviors such as a spider building a web or a whale knowing how to swim are encoded and expressed immediately upon birth without prior individual learning is a long-standing biological problem [8]. These innate behavioral capacities arise from neural circuits encoded within the genome [9].

However, Blumberg challenges the traditional notion of instinct as a preprogrammed or hardwired code solely determined by genetics. Instead, he proposes a developmental systems perspective where species-typical behaviors emerge through a complex cascade of interactions between the organism's morphology, its genome, and its "ontogenetic niche" the reliable ecological context inherited alongside genes. From this view, what appears to be a fixed instinct is actually a robust developmental outcome, where behaviors exist on a continuum from highly plastic to highly robust. [2]

Epigenetic mechanisms, which play a vital role in developmental regulation, may transform a learned behavior into an instinct by reducing its reliance on external stimuli and promoting an internally guided program of neural development [8]. These epigenetic phenomena can also give rise to traits that are heritable, enabling scientists to study how acquired characteristics persist and evolve across generations [10].

2.2 Computational Models of Evolution and Natural Selection

The foundational approach of simulating evolutionary processes, particularly using genetic algorithms (GAs), provides the context for computational evolution and artificial life. Genetic algorithms are frequently utilized to generate solutions for search and optimization problems by employing biological concepts such as crossover, mutation, and natural selection [?].

In simulations focused on protozoan-like cells, the control system permits the co-evolution of form and function. Traits are derived from the outputs of the recurrent GRN, making them potentially dynamic and subject to environmental influence or signals. In one method, evolvable traits are represented as modular components that develop from a GRN via a 'fuzzy lock-and-key' algorithm [3]. This algorithm uses the cell's complex molecules (analogous to proteins) as keys to match a node's construction signature (the lock), thereby determining which functional attachment (e.g., flagella or photoreceptor) is developed.

A key property observed in evolved GRNs is heterochrony, where small structural or concentration changes only induce small changes in network dynamics. This behavior is considered crucial for artificial life applications because it smoothes the fitness landscape, enhancing evolvability. GRNs also utilize a compact genetic encoding compared to deep neural networks, which can significantly reduce the number of variables required for optimization [4].

Artificial Gene Regulatory Networks (AGRN) model the genetic mechanisms and operations involved in regulating gene activity, allowing for phenotypic variation without modifying nucleotide sequences. AGRNs translate static genetic information into dynamic processes and behaviors. In the context of Artificial Life, AGRNs have been extensively used [?].

2.3 Development of morphology

Node functional attachments include: Flagella (motility), Spikes (damage induction), Photoreceptors (light sensitivity signals), Phagocytosis receptors (engulfing cells for feeding), and Adhesion receptors, this allows cells to bind together into multicellular structures, sharing resources and signals.

These nodes act as modular components that connect the cell’s GRN to the external environment via Input/Output (IO) channels. This modularity allows the evolutionary process to replace one function with another while utilizing pre-developed control pathways on the other side of the channel, illustrating a concept similar to “revolutionary repurposing” observed in complex biology [3].

2.4 Zero-Shot Generalization

Some research aim to achieve immediate expression of instinct-like behaviors under zero-shot conditions connects strongly with recent advancements in bio-inspired neural networks utilizing synaptic plasticity.

Hebbian Networks and Adaptation. Bio-inspired plastic neural networks, often incorporating Hebbian learning, are optimized using Evolutionary Strategies (ES). Hebbian learning allows weights to dynamically adjust based on local neural activities, a critical function in biological adaptive motor patterns

These models have demonstrated resilience to unforeseen environmental changes, such as navigating uneven terrain or coping with morphological damage (e.g., damaged legs on a robot). This capacity for immediate, dynamic self-organization upon encountering novel survival challenges strongly supports the concept of innate, adaptive capacity needed for instinct [7].

Cells are born and immediately begin the developmental process of constructing surface nodes and forming their form and function. This process is governed by the inherited GRN and its encoded traits. Since the GRN is the inherited mechanism that guides the cell’s initial construction, the resulting immediate morphology can be considered an innate, genetically encoded capacity, analogous to the innate abilities you seek to model [3].

Chapter 3 Background knowledge

This research is grounded in knowledge from multiple fields such as biology, psychology, computational evolution, and neural network. To construct a model of Instinct development. Accordingly, this chapter presents the relevant theories and prior studies under key thematic areas.

3.1 Ethology

Ethology is the scientific study of animal behavior, usually under natural conditions, with an emphasis on evolution, adaptation, and ecology.

3.1.1 Instinct and Innate Behavior

Traditionally, instinct has been defined as a genetically programmed behavior that organisms express immediately without prior learning. However, contemporary perspectives in developmental ethology, notably those of Mark S. Blumberg, [2] challenge the notion of instinct as a static, hard-wired program.

Instead, this research adopts the Developmental Systems view, defining instinct as a robust outcome of development. From this perspective, species-typical behaviors emerge from the consistent interaction between an organism's morphology, its genetic inheritance, and its reliable ecological context (ontogenetic niche). While these behaviors appear "innate" or "zero-shot" at birth, they are fundamentally the result of highly canalized developmental processes rather than direct genetic encoding.

3.1.2 Fixed action pattern

A Fixed Action Pattern (FAP) represents a specific type of instinctive behavior triggered by a sensory cue known as a "sign stimulus" or "releaser." FAPs are characterized by a neural network mechanism often referred to as an Innate Releasing Mechanism (IRM).

Key characteristics:

1. Specific stimulus
2. Stereotyped form
3. Completion once initiated

3.1.3 Baldwin Effect and Genetic Assimilation

The Baldwin Effect is an evolutionary concept that describes how an organism's capacity for learning can influence the course of biological evolution over time. The theory suggests that learned behaviors, if beneficial for survival, can create selection pressures that favor individuals with a greater innate ability to acquire those behaviors.

Over successive generations, a process known as Genetic Assimilation occurs. Traits that initially required extensive trial and error learning become increasingly innate (genetically canalized) as the population evolves to reduce the dependency on environmental stimuli. This mechanism explains how complex survival instincts can evolve from what were originally learned adaptations.

3.2 Evolution mechanism

Evolutionary change is driven by several fundamental mechanisms that alter the genetic composition of a population over time.

3.2.1 Natural selection

Natural selection is the process by which traits that help organisms survive and reproduce become more common in a population over generations. Individuals with traits that confer a survival or reproductive advantage (higher fitness) are more likely to pass their genes to the next generation, gradually shaping the population's characteristics.

3.2.2 Mutation

Mutation is the primary source of genetic variation, particularly in asexual populations. It involves random alterations in the DNA sequence. While many mutations may be neutral or harmful, beneficial mutations provide the raw material upon which natural selection acts, allowing organisms to adapt to new environmental pressures.

3.3 Heredity

Heredity is a biological inheritance. It is a passing on of behavior from the parent to their child. Either in sexual or asexual reproduction the child can get their parent genetic information. The passed on genetic information can be cause of variant in species to evolve.

3.3.1 Genetic Inheritance

This refers to the transmission of the stable genetic code (DNA), which determines the organism's fundamental structure and potential. In the context of this research, this corresponds to the transmission of the Gene Regulatory Network's topology and initial parameters.

3.3.2 Epigenetic Inheritance

Epigenetics involves heritable changes in gene expression that do not involve alterations in the DNA sequence. These mechanisms allow environmental factors and life experiences to influence how genes are regulated (activated or silenced). Crucially, epigenetic states can be transmitted across generations, enabling offspring to inherit "physiological states" or "pre-adaptations" from their parents, serving as a rapid response mechanism to environmental change.

3.4 Evolutionary computational

Evolutionary computational is a computer science method or an algorithm to global optimization. The idea is inspired from biological evolution. It is a population-based trial and error. When a new generation is produced it has small changes, mixing parental information. As a result, the population converges into the fitness solution.

3.4.1 Artificial life

Artificial life (A-life) refers to the field of study and creation of systems that exhibit properties characteristic of biological life, but are instantiated through alternative substrates such as computational models, robotic platforms, or synthetic chemical processes. Rather than replicating life at the molecular level, artificial life seeks to explore the fundamental principles that underlie living systems, including self-organization, adaptation, reproduction, and evolution.

3.4.2 Genetic Algorithm

A Genetic Algorithm (GA) is an evolutionary computation technique inspired by the theory of natural selection. It simulates the process of biological evolution to find optimal solutions of complex problems. The key idea is to treat each possible solution as an individual within a population and allow this population to evolve over successive generations through mechanisms that mimic the natural genetics, such as selection, crossover, and mutation.

The process begins with a randomly generated set of potential solutions (or Gene). Each gene is evaluated using a fitness function, which measures how well it solves the problem. Genes with higher fitness are more likely to be selected for reproduction. During reproduction, selected genes undergo crossover and mutation to create new solutions. This iterative process continues across many generations, improving the population's overall fitness until an optimal solution is found.

3.4.3 Gene regulation network

The gene regulatory network (GRN) is like a neural network. It regulates gene expression by determining which genes are activated or deactivated under each condition. The connections between genes have adjustable "weights," allowing the network to adaptively modify gene interactions in response to internal states or environmental stimuli. This mechanism supports the emergence of complex and flexible behaviors in organisms.

3.4.4 Synaptic Plasticity and Hebbian Learning

Hebbian Theory proposed by Donald Hebb, this theory is often summarized as "Cells that fire together, wire together." It postulates that the synaptic strength between two neurons increases if they are activated simultaneously.

Role in this Research. The context of the GRN controller, Hebbian learning serves as the mechanism for Phenotypic Plasticity. It allows the agent to adjust its internal wiring based on sensory experiences during its lifetime. This plasticity is the driving force that allows the Baldwin Effect to operate, acting as a temporary "scaffold" for survival before genetic evolution catches up.

Chapter 4 Methodology

This chapter presents the methodology developed to design, implement, and evaluate a computational model that simulates instinctive survival behaviors in artificial organisms. The proposed framework draws inspiration from biological processes, combining principles from evolutionary computation, artificial life, and developmental biology to capture the adaptive and emergent properties of living systems.

At the core of this approach is a genetic encoding mechanism, which defines the heritable traits and structural configurations of the artificial organisms. These genetic blueprints are expressed through gene regulatory networks (GRNs), dynamic systems that govern the interaction between genes and regulate the development and behavioral responses of the agents. The GRNs enable the emergence of complex, context-dependent behaviors, allowing artificial organisms to adapt to environmental challenges in a manner analogous to biological evolution.

The methodology integrates online evolution, where adaptation occurs continuously during the agent’s lifetime, with evolutionary selection, which operates across generations to refine successful traits. This dual-layered process ensures that both immediate behavioral adaptation and long-term evolutionary optimization contribute to the agents’ survival strategies. Through this integration, the model aims to bridge the gap between low-level genetic processes and high-level behavioral phenomena, providing insights into how instinctive survival mechanisms can emerge in artificial systems.

4.1 Conceptual

4.1.1 Emergence of instinct in agent

Emergence of Instinct via Evolutionary Algorithms In this simulation, agent behaviors are not pre-programmed. Instead, they emerge from the dynamics of a Gene Regulatory Network (GRN) that functions as the agent’s neural controller. We simulate the process of natural selection, where agents capable of survival and resource acquisition have a higher probability of reproducing.

The methodology focuses on the evolution of developmental parameters rather than fixed behaviors. Agents inherit a genome encoding initial connection weights, synaptic plasticity rates, and epigenetic states. Over successive generations, the evolutionary pressure selects for genotypes that facilitate rapid and robust behavioral development. This process simulates the Baldwin Effect, where initially plastic, learned behaviors become genetically assimilated, allowing descendant agents to exhibit effective ‘instinctive’ survival strategies immediately upon initialization (zero-shot performance) without the need for extensive prior learning.

4.1.2 Gene Inheritance

When an agent meets the necessary conditions for reproduction such as possessing sufficient energy or residing within an environment conducive to reproductive activity. It becomes eligible to transmit its genetic information to the next generation. In this study, reproduction is modeled as an asexual process, meaning that each offspring inherits its DNA directly from a single parent organism. To introduce genetic diversity within the population, a mutation mechanism is applied during replication. This process ensures the emergence of genetic variations, which play a crucial role in promoting adaptability and evolutionary dynamics among artificial organisms.

4.1.3 Natural selection and evolution process

This study is based on the principles of natural selection and evolution, which form the foundation for adaptive behavior and diversity in artificial organisms. In nature, natural selection drives evolution by favoring individuals with genetic and behavioral traits that improve their chances of survival and reproduction. These beneficial traits are passed to the next generation, while less effective traits gradually disappear. Over many generations, populations become better adapted to their environments.

In the computational model, the evolutionary process takes place over repeated generations through these mechanisms include reproduction, mutation, and selection. Reproduction transfers genetic material to new agents, mutation introduces small random changes, and selection allow the fittest agents to pass on their genes. Through this cycle, artificial organisms gradually develop more complex behaviors and stronger survival strategies. By using these mechanisms, the model simulates how evolution and natural selection create adaptive behavior, providing a simplified but effective analogy to biological evolution.

4.1.4 Energy based concept

In this study, energy functions as the primary resource essential for an agent's survival, growth, and maintenance of biological processes. The ability to acquire and regulate energy is a fundamental property of life and a key determinant of an agent's persistence within the environment. The simulation is designed as an energy based system, meaning that every activity ranging from basic actions such as movement and rotation to complex processes like constructing new organelles requires the cost of energy. Agents must therefore continuously gather and manage energy to sustain their operations and ensure survival. Within the simulation, energy can be obtained from three main sources: plant cells which serve as the primary energy providers, dead cells which originate from other agents that have dead and photosynthesis that agent need to construct a chloroplast to obtain energy. These energy sources sustain the agents' activities and directly influence their behavior, competition, and evolutionary success.

4.2 System overview

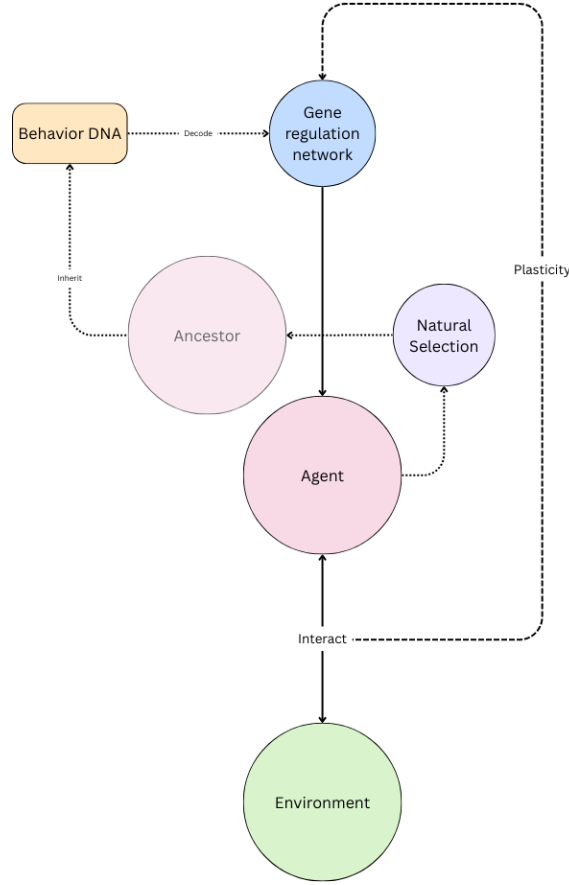


Figure 4.1: Overview of agent and environment

The proposed evolution framework is founded on two core principles. (1) Online evolution, enabling the agent to evolve while operating in real time, and (2) Evolution by natural selection, inspired by biological evolutionary processes.

Online evolution refers to the adaptive process through which an agent modifies its internal genetic representation and regulatory networks during ongoing interaction with the environment. Unlike traditional evolutionary approaches that occur across generations, online evolution enables adaptation within an agent’s lifetime. This mechanism allows genetic and network-level adjustments in response to environmental stimuli, thereby enhancing the agent’s ability to deal with and thrive in dynamic environments.

Natural selection process operates through the differential survival and reproduction of individuals based on variations in their genetic and behavioral traits. Agents that exhibit characteristics better suited to their environment have a higher probability of reproducing and transmitting their genetic to subsequent generations. Consequently, genes that confer adaptive advantages become more prevalent and are expressed more strongly within the population, while those that are inefficient gradually diminish over time.

4.3 Agent

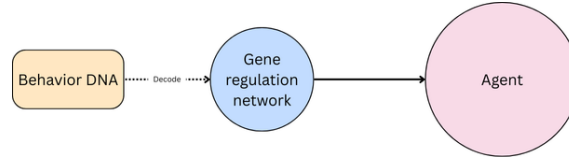


Figure 4.2: Agent

In this model, each agent is composed from a DNA, that defines the agent’s behavior, which governs its actions and responses.

Each agent has a group of internal parameters that describe its physiological and functional state. These parameters include Speed which determines movement capability, Health points which represent physical condition, Age indicating the life stage, Energy reflecting available resources, Size, defining physical scale and Mutation rate, which influences the likelihood of genetic variation during reproduction.

4.3.1 Behavior

The behavior of each agent is controlled by an internal network composed of interconnected nodes. In this framework, there are three types of nodes: Input (sensory) nodes, Action nodes, and Special nodes.

The Input nodes function as the agent’s sensory system, receiving signals from the environment such as light intensity or other external stimuli as well as internal parameters that describe the agent’s state.

Action nodes represent the set of basic actions available to the agent, such as moving forward, rotating, or activating specialized organelles (e.g., using a flagellum to increase speed if the agent possesses one).

The Special nodes are designed to introduce behavioral variation and complexity. These include internal processing nodes that enable the agent to produce different actions in response to the same input, as well as oscillator or randomization nodes that generate rhythmic or stochastic variations in behavior. Together, these components allow agents to display adaptive and diverse behavioral patterns in dynamic environments.

The nodes and connection weights that define an agent’s behavior are determined by its DNA, which serves as a genetic blueprint for the internal control network. The DNA encodes both the structure of the network specifying how nodes are connected and the initial weights of these connections. When the DNA is decoded, it reveals the regulatory relationships that shape how the agent perceives inputs and produces actions. In this framework, the DNA works in coordination with the gene regulatory network (GRN), ensuring that genetic information dynamically influences behavioral development and adaptation to the environment. The detailed structure and architecture of the DNA, including its encoding scheme and representation of network parameters, will be explained in the following section.

4.3.1.1 DNA structure

$$G_N = \langle Type, ID, \beta, \tau, \mathcal{E} \rangle \quad (4.1)$$

$$G_C = \langle Type, ID_{src}, ID_{tgt}, W, \eta, S \rangle \quad (4.2)$$

Where:

- $Type$: Gene Type (Node or Connection)
- β : Innate Bias
- τ : Time Constant
- $\mathcal{E}$: Expression Level
- W : Initial Weight
- η : Learning Rate
- S : State

In the proposed framework, the DNA is represented in hexadecimal format, where each block consists of six hexadecimal digits. Each block can be decoded to identify which nodes are connected and the weight associated with each connection in the agent's internal network. The length of an agent's DNA varies depending on its inherited characteristics and its ability to gather energy from the environment. Agents with longer DNA sequences typically possess more complex structures and behaviors, but they also require greater energy to sustain themselves. Consequently, agents must evolve strategies to balance the complexity of their DNA with the energy needed for survival, promoting efficient adaptation within the system.

To optimize storage efficiency, each DNA parameter is encoded in a hexadecimal format, allowing for direct decoding into a functional Gene Regulatory Network (GRN). The proposed genetic architecture is distinctively segmented into Node Genes and Connection Genes. Node Genes are responsible for selecting functional units from a predefined pool, thereby establishing the coarse network topology, while Connection Genes define the fine-grained topology by specifying the wiring and interaction dynamics between these nodes.

4.3.1.2 Network update mechanism

The behavior of each node within the Gene Regulatory Network (GRN) is governed by a set of differential equations approximated for discrete-time simulation. Let $y_i(t)$ represent the internal activation state of node i , and $X_i(t)$ represent its final output signal. The network updates its state iteratively using the following equations

First, the target activation for node i is calculated by aggregating all incoming signals from presynaptic nodes j , multiplied by their respective synaptic weights W_{ji} , plus the node's innate bias β_i

$$Target_i(t) = \sum W_{ij}(t)X(t) + \beta_i \quad (4.3)$$

Next, the internal activation state y_i is updated over a time step Δt . The rate of this change is modulated by the time constant τ_i , which dictates the temporal inertia of the node

$$y_{(t+\Delta t)} = y_i(t) + \frac{\Delta t}{\tau_i} (Target_i(t) - y_i(t)) \quad (4.4)$$

Finally, the output signal X_i is generated by applying a hyperbolic tangent (tanh) activation function. Crucially, the internal state is scaled by the epigenetic expression level $\mathcal{E} \in [0, 1]$ before the non-linearity is applied, acting as a genetic gain control mechanism

$$X_i(t + \Delta t) = \tanh(\mathcal{E}_i \times y_i(t + \Delta t)) \quad (4.5)$$

Where

- y_t : Activation of the node
- *Target*: The target activation value derived from incoming signals.
- τ : Time Constant, the rate of response (temporal inertia).
- W_i : The synaptic weight of the connection from node i .
- X_i : The output value of the presynaptic node i .
- β : The innate bias of the node.
- *Expression*: The epigenetic expression level, acting as a sensitivity gain.

In summary, while the Weights determine the connectivity structure, the Time Constant determines the temporal dynamics, and the Expression Level determines the physiological state. Together, these parameters allow the GRN to function not just as a reactive controller, but as a dynamic developmental system capable of exhibiting the complex, robust behaviors defined as instinct.

4.3.1.3 Synaptic Plasticity and Hebbian Learning

To enable lifetime adaptation within the Gene Regulatory Network (GRN), this research incorporates a mechanism of Synaptic Plasticity based on Hebbian Theory. Donald Hebb's postulate, often summarized as "Cells that fire together, wire together," suggests that the synaptic efficacy between two neurons increases if they are persistently active simultaneously.

The change in synaptic weight ΔW_{ij} is driven by the correlation of internal activations between the connected nodes, scaled by the genetically encoded learning rate η_{ij}

$$\Delta W_{ij}(t) = \eta_{ji} \cdot (y_j(t) \cdot y_i(t)) \quad (4.6)$$

$$(4.7)$$

To ensure network stability and prevent unbounded weight growth (weight explosion), a passive synaptic decay (forgetting factor) denoted as λ is introduced. The weight is updated for the next time step as follow

$$W_{ji}(t + \Delta t) = W_{ji}(t) + \Delta W_{ji}(t) - (\lambda \cdot W_{ji}(t) \cdot \Delta t) \quad (4.8)$$

Where λ is a small constant representing the metabolic degradation of synaptic strength over the time, and the final weight is bounded within a biologically plausible range $[-2.0, 2.0]$.

This mechanism serves as the driver for Phenotypic Plasticity. By encoding the Learning Rate (η) in the genome, the evolutionary process can control how much a specific behavior should be learned versus how much should be innate.

This interaction between genetically defined plasticity and lifetime experience is the fundamental component enabling the simulation of the Baldwin Effect.

4.3.1.4 The Metabolic Cost of Plasticity

To effectively simulate the Baldwin Effect and drive the process of genetic assimilation, lifetime learning must incur a biological penalty. If phenotypic plasticity is strictly cost-free, the evolutionary process will not generate the necessary selection pressure to transition learned behaviors into innate instincts. Agents would simply remain highly plastic indefinitely without any evolutionary disadvantage.

Therefore, this computational model introduces a Metabolic Cost of Plasticity. Every structural modification within the GRN (Gene Regulatory Network) specifically, the adjustment of synaptic weights requires a continuous expenditure of the agent's internal energy. The energy consumed by plasticity at any given time step t is directly proportional to the absolute magnitude of the synaptic changes occurring across all active connections.

Let $c_{plasticity}$ denote a constant metabolic penalty factor that scales the severity of the learning cost. The total plasticity cost $C_p(t)$ incurred by an organism per time step is defined as

$$C_p(t) = c_{plasticity} \sum_{j,i} |\Delta W_{ji}(t)| \quad (4.9)$$

This calculated cost is continuously subtracted from the agent's overall energy reserves. Consequently, artificial organisms that rely heavily on high learning rates (η) to adapt to their environment will experience rapid energy depletion. If an agent

spends excessive energy rewiring its brain, it risks starvation and failure to reach the reproduction threshold.

This mechanism creates a robust natural selection pressure. Evolution favors individuals that inherit optimal initial synaptic weights (W_{start}) that are already aligned with survival strategies, thereby requiring minimal lifetime modification. Over successive generations, this dynamic forces the population’s average learning rate to decrease while survival rates increase, successfully demonstrating the genetic assimilation of adaptive behaviors into zero-shot survival instincts.

4.4 Evolutionary process

The evolutionary process in this research is modeled after Natural Selection integrated with Developmental Systems Theory. It employs an asexual reproduction scheme to rigorously trace the lineage of behavioral traits and epigenetic states.

4.4.1 Asynchronous Evolutionary

The asynchronous evolutionary loop is designed to handle continuous evaluation, real time developmental dynamics, and implicit fitness. Algorithm 1 describes the life cycle of an artificial agent, explicitly highlighting the sequence of sensory integration, deterministic brain updates, and the critical application of the metabolic plasticity cost.

Algorithm 1: Asynchronous Evolutionary with Plasticity Cost

```
1 Function UpdateAgent(agent, dt):
2   agent.age  $\leftarrow$  agent.age + dt
3   agent.energy  $\leftarrow$  agent.energy - (MetabolismCost  $\times$  dt)
4   agent.FeedInputs()
5   agent.brain.TickBrain(dt)
6     cost_p  $\leftarrow$  agent.brain.currentFramePlasticityCost
7   if cost_p > 0 then
8     agent.energy  $\leftarrow$  agent.energy - cost_p
9     agent.lifetime_learning_cost  $\leftarrow$ 
10      agent.lifetime_learning_cost + cost_p
11
12   agent.ExecuteActions()
13
14   /* Natural Selection */
15   if agent.energy  $\leq$  0 or agent.age > MaxLifespan then
16     World.Remove(agent)
17     return
18
19   /* Reproduction */
20   if agent.energy  $\geq$  ReproductionThreshold then
21     offspring  $\leftarrow$  Reproduce(agent)
22     World.Add(offspring)
23
24 Function Reproduce(parent):
25   parent.energy  $\leftarrow$  parent.energy - ReproductionThreshold
26   offspring  $\leftarrow$  Clone(parent)
27   /* Inherit current epigenetic state */
28   offspring.dna.expression  $\leftarrow$  parent.brain.current_expression
29   /* Reset plasticity cost to prevent inherited debt */
30   offspring.lifetime_learning_cost  $\leftarrow$  0
31   /* Apply Variation */
32   offspring  $\leftarrow$  Mutate(offspring)
33   return offspring
```

Unlike traditional Genetic Algorithms that operate in discrete generations where the entire population is evaluated and replaced simultaneously, this research employs an Asynchronous Evolutionary Model (also known as a Continuous or Steady-State System). In this framework, generations overlap, and evolutionary events occur independently for each agent based on its interaction with the environment.

The process is driven by Implicit Fitness, where survival and reproduction are not determined by an external function, but by the agent's ability to maintain its internal energy balance and ability to reproduction.

The process follows a continuous simulation loop:

1. Population Initialization: The simulation begins by instantiating an initial population (Generation 0) of artificial agents. Each agent is initialized with a randomly generated synthetic genome.
2. Continuous Time Loop: At every simulation time step (t), each agent operates independently.
3. Selection: The agent that can survive (fitness), Reproduction is an energy-dependent event. If an agent maintains an energy level above a specific Reproduction Threshold, it triggers an asexual reproduction process.
4. Mutation Strategy: Immediately after cloning, the offspring’s genome is subjected to stochastic mutations to introduce variation.

4.4.2 Mutation

Mutation Strategy To drive evolutionary adaptation and facilitate the Baldwin Effect, the mutation process (Algorithm 2) applies stochastic variations to the offspring’s genome across three distinct levels: epigenetic state, parametric properties, and network topology.

1. Value Mutation: This process fine-tunes the existing properties of the network without altering its shape.
 - Epigenetic Drift: Low-level noise is applied to the inherited expression levels (E). This prevents the population from stagnating in local epigenetic optima and simulates biological noise in gene regulation.
 - Parametric Mutation: The node’s intrinsic properties (bias and σ) and the connection properties (initial weight and learning rate η) are stochastically adjusted. Mutating η allows evolution to tune the temporal dynamics, while mutating the learning rate η explicitly controls the degree of phenotypic plasticity—the core driver of the Baldwin Effect.
 - Gene Toggling: Existing connections have a small probability of being switched on or off, acting as a rapid, reversible evolutionary switch.
2. Structural Mutation: To allow for the complexification and simplification of the agent’s neural controller, topological mutations alter the network’s architecture.
 - Constructive Mutation: New nodes (either standard inputs/actions or hidden nodes) and new synaptic connections can be randomly added to the network, increasing behavioral capacity.
 - Destructive Mutation: To counteract network bloat and minimize the metabolic cost of maintaining useless synapses, nodes and connections can be permanently removed. Crucially, when a node is deleted, a cleanup mechanism automatically prunes any orphaned connections (synapses where the source or target was the deleted node), ensuring the integrity of the Gene Regulatory Network.

Algorithm 2: DNA Mutation

```
1 Function MutateDNA(child_dna,  $\mu$ ):  
   /* Value Mutation */  
2   foreach node  $\in$  child_dna.nodes do  
3     if rand()  $< \mu$  then  
4       node.bias  $\leftarrow$  MutateGene(node.bias)  
5     if rand()  $< \mu$  then  
6       node.tau  $\leftarrow$  MutateGene(node.tau)  
7     if rand()  $< \mu$  then  
8       node.expression  $\leftarrow$  MutateGene(node.expression)  
       // Epigenetic drift  
9   foreach conn  $\in$  child_dna.connections do  
10    if rand()  $< \mu$  then  
11      conn.weight  $\leftarrow$  MutateGene(conn.weight)  
12    if rand()  $< \mu$  then  
13      conn.learningRate  $\leftarrow$  MutateGene(conn.learningRate)  
14    if rand()  $< \mu \times 0.2$  then  
15      conn.state  $\leftarrow$  not conn.state           // Toggle gene state  
  
   /* Structural Mutation (Topology) */  
16  if rand()  $< \mu \times 0.5$  then  
17    AddConnection(child_dna)  
18  if rand()  $< \mu \times 0.2$  then  
19    AddNode(child_dna)  
20  if rand()  $< \mu \times 0.5$  then  
21    RemoveConnection(child_dna)  
22  if rand()  $< \mu \times 0.2$  then  
23    deleted_id  $\leftarrow$  RemoveNode(child_dna)  
24    child_dna.connections.RemoveAll(c  $\rightarrow$  c.src ==  
    deleted_id or c.tgt == deleted_id)  
25  return child_dna
```

4.5 Environment

The environment used in this study is a two-dimensional (2D) aquatic world, designed to simulate a dynamic and interactive ecosystem in which agents can move, gather energy, and evolve.

4.5.1 Source of Energy

In this 2D environment, energy is physically instantiated as consumable resources scattered throughout the spatial domain. The simulation enforces a closed loop ecosystem (The Circle of Life) where energy is continuously recycled. There are two primary sources of energy

1. Environmental Nutrients: Energy pellets are systematically generated by a spawner within a bounded ecosystem. To prevent infinite resource accumulation, the environment maintains a strict carrying capacity (maximum food count).
2. Biological Corpses: When an artificial agent dies either from energy depletion (starvation) or reaching its maximum biological lifespan its remaining biomass is immediately converted back into an energy pellet at its exact location of death. This mechanism allows scavenger behaviors to evolve, where agents adapt to trail older or weaker agents to consume their remnants.

4.5.2 Spatial Interaction and Sensory Mechanics

Agents interact with the environment through biologically inspired simulated physics. The sensory and physical boundaries are rigidly defined to enforce spatial awareness

1. Vision (Raycasting): Agents perceive the environment using simulated bilateral vision (Left and Right Eyes). This is implemented via 2D Raycasting, projecting sensory vectors at specific angles. The visual signal is inversely proportional to the distance of the detected food within a predefined sight range. If food is close, the sensory node outputs a signal approaching 1.0 if nothing is detected, the output is 0.0.
2. Eating Mechanism: Resource consumption requires precise spatial alignment. When the GRN triggers the Eat Action node, the agent expends an energy to activate an Area of Effect overlap circle defined by its eatRadius. Only food entities physically colliding with this specific radius during the bite action are successfully consumed and converted into internal energy.

4.5.3 Dynamic environment

To prevent the artificial organisms from converging on overly simplistic, static survival strategies (such as remaining stationary in a food rich area), the simulation environment is designed to be dynamic. The ecosystem continuously introduces environmental variations that compel the agents to adapt

1. Resource Fluctuations: The availability of environmental nutrients is not strictly constant. Parameters such as the food generation rate and the ecosystem's carrying capacity can fluctuate over time. This mimics seasonal changes or periods of ecological scarcity, rigorously testing the population's metabolic efficiency and forcing evolutionary shifts between highly active hunting and energy conserving hibernation strategies.
2. Spatial Constraints: Physical barriers and walls can be introduced or dynamically shifted to divide the ecosystem into distinct zones. This geographical complexity forces agents to evolve efficient navigation, spatial memory, and obstacle avoidance behaviors utilizing their stereoscopic raycast vision.

3. Thermal Gradients and Light Intensity (Next Phases): The environmental architecture is built to support varying abiotic factors like temperature and light zones. For instance, extreme thermal zones will directly penalize an agent by exponentially increasing its basalMetabolism or directly damaging its Health Points. This will drive the evolution of thermoregulatory migration forcing agents to utilize sensory nodes to seek optimal comfort zones.

4.5.4 Survival regulation

Survival regulation refers to a set of strict biological constraints that govern the agents' life processes within the environment. Agents operate under a continuous energy deficit and must actively scavenge or hunt to survive. The energy dynamics and life cycles are strictly regulated by the following metabolic penalties and rules.

1. Basal Metabolism: A constant rate of energy decay applied every time step merely for existing, simulating the fundamental cost of biological maintenance.
2. Kinetic Cost: Physical actions initiated by the Gene Regulatory Network incur specific energy penalties. For instance, moving forward or rotating consumes additional energy proportional to the action's duration and intensity. This forces agents to evolve efficient movement trajectories rather than moving erratically.
3. The Cost of Plasticity: As previously defined, altering synaptic weights during lifetime learning incurs a heavy metabolic tax, which serves as the primary evolutionary pressure for the Baldwin Effect.
4. Aging and Mortality: Agents possess a genetically defined maximum lifespan. Regardless of their accumulated energy reserves, once their chronological age exceeds this limit, programmed cell death is triggered. This ensures generational turnover and prevents functionally immortal agents from dominating the ecosystem.
5. Reproductive Constraints: Asexual reproduction is constrained by both maturity and energy. An agent must reach a minimum maturity age and accumulate energy beyond a specific Reproduction Threshold. Upon reproducing, a significant portion of the parent's energy is directly consumed to synthesize the offspring's initial energy and construct its neural network, effectively resetting the parent's energy level and preventing infinite instant cloning.

Examples of survival regulations include energy depletion, where agents die if their energy drops below a critical threshold. Aging, which limits the lifespan of each agent. By incorporating these constraints, the simulation produces more realistic evolutionary behavior, promoting the development of adaptive strategies and instinct like responses among agents.

4.6 Simulation implementation

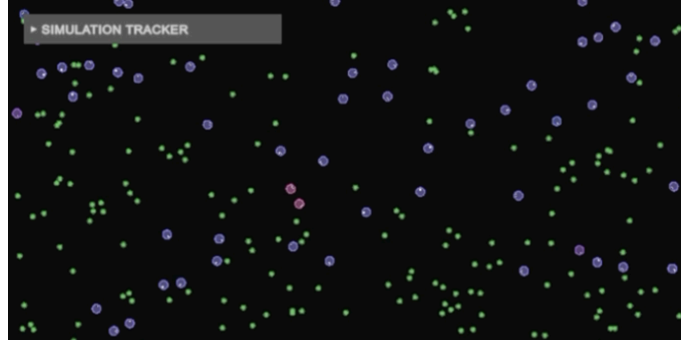


Figure 4.3: simulation screenshot

The simulation in this study is developed using the Unity engine, a versatile game development platform that provides real-time rendering, physics simulation, and customizable environmental control. Unity’s flexibility allows for efficient implementation of complex agent behaviors, environmental dynamics, and visualization of evolutionary processes in an interactive two-dimensional world.

4.7 Experiment design

1. **Evolutionary Trajectory Analysis.** The simulation runs for a prolonged period in a static environment. We analyze the correlation between Fitness and Average Plasticity Index. The hypothesis is that as fitness maximizes, plasticity will minimize.
2. **Epigenetic Resilience.** The environment is dynamically altered. The survival rates of agents with enabled Epigenetic Inheritance are compared against a control group (Genetic only) to demonstrate the benefits of transgenerational state transmission.
3. **Zero-shot Validation.** High-fitness agents from the final generation are instantiated with their learning mechanisms disabled ($\eta = 0$). Their survival rate is compared against naive agents to validate the of instinct.

4.8 Evaluation

To rigorously evaluate the simulation outcomes and validate the occurrence of the Baldwin Effect and genetic assimilation, the tracking module continuously logs the following quantitative metrics

4.8.1 Average Learning Rate

This metric tracks the evolutionary trajectory of the population’s reliance on phenotypic plasticity. It is calculated as the mean genetically encoded learning rate across all active synapses in the population at any given time. A progressive decrease in over generations, concurrent with high survival rates, serves as primary evidence

that behaviors are transitioning from heavily plastic (learned) to robustly innate (instinctive).

4.8.2 Average Lifetime Plasticity Cost

This metric quantifies the metabolic burden of learning. It aggregates the total energy expended on synaptic modifications over each agent’s entire lifespan, subsequently averaged across the current population. A successful demonstration of the Baldwin Effect will exhibit a distinct, two phase trend, an initial spike in this cost (as early generations rely heavily on Hebbian learning to survive) followed by a gradual, permanent decline (as natural selection assimilates the adaptive traits into the genome, reducing the need for costly lifetime learning).

4.8.3 Average Plasticity Index

This metric tracks the evolutionary trajectory of the population’s reliance on learning. It is calculated as the mean learning rate across all connections in the population. A decrease in $\bar{\eta}$ over time, concurrent with increasing fitness, serves as primary evidence for Genetic Assimilation, indicating that behaviors are transitioning from plastic to innate instinctive.

4.8.4 Innate weight Convergence

This metric measures the diversity of initial weights (W_{start}) within the gene pool. It is calculated using the standard deviation of innate weights for critical connections. High variance indicates a random search phase, while low variance (convergence) indicates that evolution has discovered an optimal "innate bias" for behavior.

4.8.5 Zero-shot Performance

In the test scenario, the agent should demonstrate survival behavior immediately or almost immediately after birth, without requiring additional training or minimal adaptation time. The exhibited behavior is expected to reflect inherited instincts derived from its ancestors. Even when the environment undergoes slight changes, the agent should be capable of adapting to the new conditions and continuing to display effective survival behaviors.

4.8.6 Gene backtracking

Beyond numerical metrics, the simulation utilizes a serialized data structure to periodically capture and store the complete DNA strings of the population. This allows for phylogenetic backtracking enabling researchers to reconstruct an agent’s ancestral lineage, instantiate historical generations in an isolated environment, and perform comparative analyses on how specific mutations and topological shifts contributed to the emergence of instinctive behaviors over time.

Chapter 5 Experiments and Results

5.1 Ecosystem Equilibrium and Population Dynamics

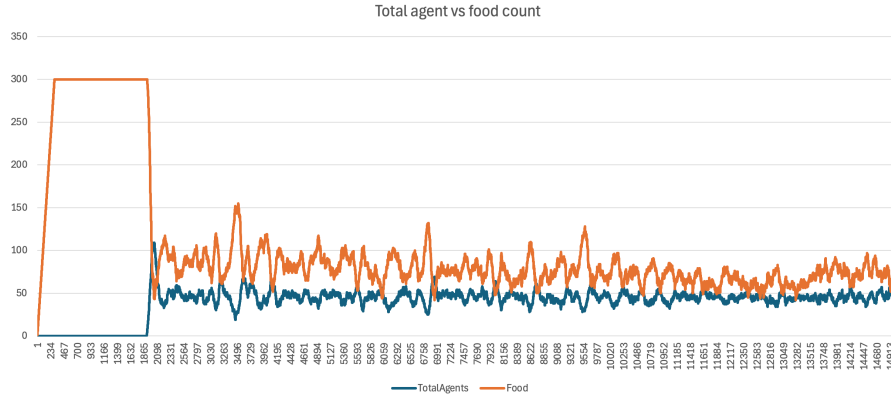


Figure 5.1: Total Agent vs food count

The simulation was initialized by allowing the food to reach the maximum carrying capacity of 300 units. Upon the introduction of the initial generation (at timestep 1865), a rapid consumption phase occurred, causing a sharp decline in available food.

Following this initial colonization phase, the ecosystem successfully transitioned into a dynamic equilibrium. The total agent population stabilized around an average of 40-60 individuals, while the food count fluctuated rhythmically between 60 and 120 units. This Lotka-Volterra like oscillation proves that the implemented survival regulations specifically basal metabolism, aging, and reproductive energy thresholds functioned correctly to prevent unbounded population explosion and ecological collapse. The artificial organisms demonstrated sustainable foraging behaviors, maintaining a stable balance with their renewable environment.

5.2 Genetic Diversity

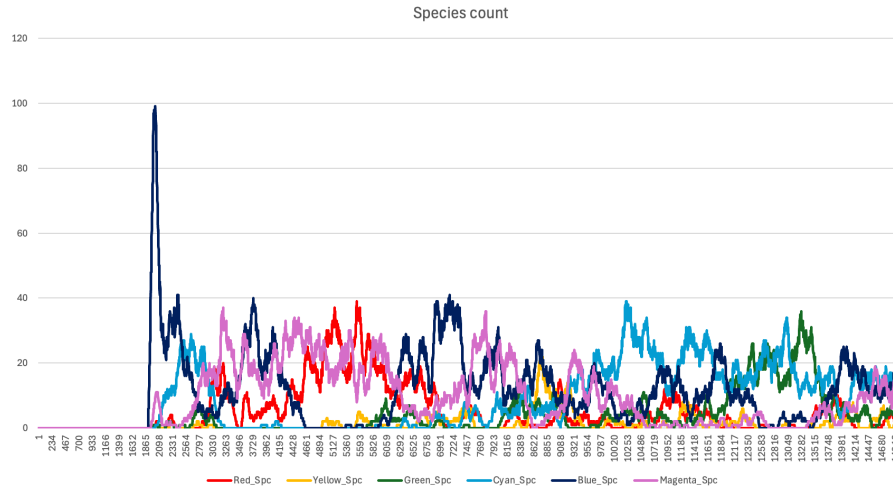


Figure 5.2: Species count

To analyze morphological and genetic variation, the simulation tracked the population distribution across distinct "species," categorized by the Gene Regulatory Network's topological Center of Mass (color coded). The results illustrate a highly dynamic, continuous evolutionary process.

Initially, a single dominant genetic cluster (Blue species) monopolized the ecosystem. However, due to structural and parametric mutations, this dominance was transient. Over time, the ecosystem witnessed continuous species succession the Blue species declined as Cyan, Magenta, Green and Red species emerged and competed for dominance. The absence of a permanent, single dominant species indicates that the fitness landscape is constantly shifting. The artificial lifeforms continuously evolved diverse topological strategies to optimize energy acquisition, ensuring long-term genetic diversity.

5.3 The Emergence of the Baldwin Effect

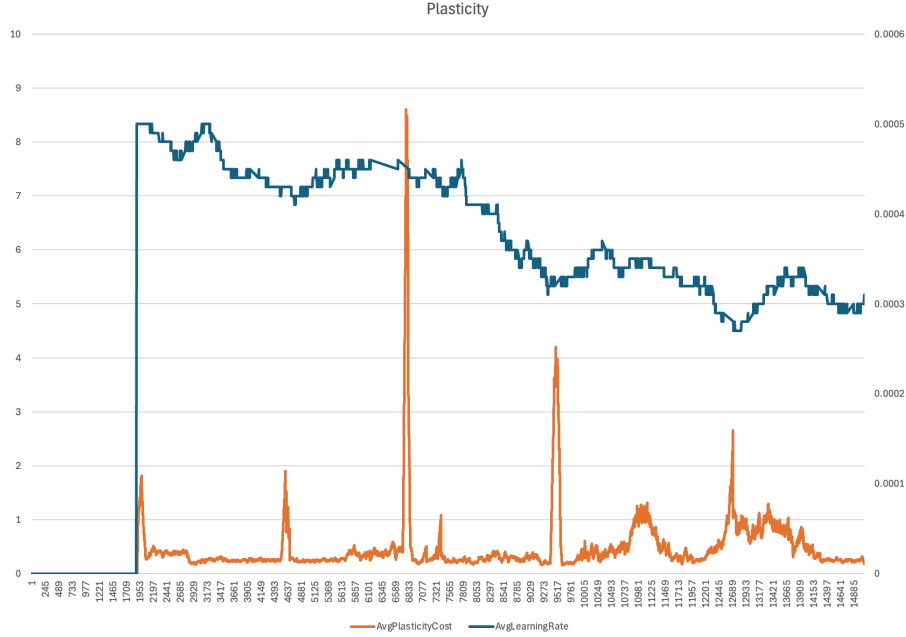


Figure 5.3: Plasticity

The primary objective of this research was to observe the Baldwin Effect the genetic assimilation of learned behaviors into innate instincts driven by the metabolic cost of plasticity. The recorded Average Plasticity Cost metric provides empirical evidence of this phenomenon.

Unlike an unconstrained system where learning cost accumulates indefinitely, the corrected generational inheritance ensures that offspring are born with zero accumulated plasticity debt. The graph demonstrates that the baseline Average Plasticity Cost remained remarkably low and stable (averaging below 1.0) throughout the majority of the simulation. This low baseline strongly suggests that the population converged upon highly optimized innate weights and low learning rates (η) that required minimal lifetime rewiring to survive.

Crucially, the graph features several massive spikes (e.g. reaching 8.6 at timestep 6758 and 4.2 at 9554). These spikes represent the emergence of highly plastic mutant generations that attempted to adapt via intense Hebbian learning. However, the rapid, immediate crash of these spikes back to the baseline proves the efficacy of the Metabolic Cost of Plasticity. Organisms that expended too much energy were swiftly penalized by starvation and outcompeted by agents possessing pre optimized innate instincts. Thus, natural selection actively pruned highly plastic individuals, favoring zero shot instinctual responses and validating the computational model of the Baldwin Effect.

And as illustrated in the dual axis Plasticity chart, the correlation between the Average Learning Rate (blue line) and the Average Plasticity Cost (orange line) provides compelling evidence of the Baldwin Effect and Genetic Assimilation.

Upon the introduction of the initial randomized population (around time step 1953), the Average Learning Rate immediately peaked near 0.0005. This indicates that the early generations, lacking robust innate instincts, were forced to rely heavily on phenotypic plasticity (Hebbian learning) to adapt their internal neural weights and survive in the environment.

The simulation recorded severe spikes in the Plasticity Cost, notably at timesteps 6833 and 9517. These spikes represent generations with excessively high learning rates expending massive amounts of metabolic energy to rewire their brains. The subsequent rapid collapse of these spikes confirms the efficacy of the implemented survival regulations organisms that incurred excessive metabolic costs for learning were swiftly outcompeted or starved, demonstrating a strong evolutionary pressure against over-reliance on lifetime plasticity.

The most critical observation occurs post timestep 10000. Despite maintaining a stable population and foraging effectively, the population's Average Learning Rate exhibits a distinct, continuous downward trend, declining from its peak of 0.0005 to approximately 0.0003 by the end of the recorded period. This macro-evolutionary shift confirms that the simulated natural selection favored agents that inherited optimized innate synaptic weights. Consequently, traits that originally required costly lifetime learning were genetically assimilated, allowing descendant agents to execute survival behaviors instinctively with significantly reduced plasticity rates.

Chapter 6 Timeline

No	Tasks	2025				2026				
		Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May
1	Literature review									
2	Design Entities architecture									
3	Submit Pre-Proposal									
4	Design genetic structure									
5	Present Proposal									
6	Develop simulation program									
7	Present progress update 1									
8	Develop GRN framework									
9	Present progress update 2									
10	Prepare report									
11	Final thesis defense									

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